Ex.No. 12: NB-IoT vs LTE-M Throughput Simulation Date

AIM: To simulate and compare the performance of NB-IoT and LTE-M communication technologies in terms of Throughput (Mbps), Latency (ms), and Packet Delivery Ratio (%) using MATLAB.

Objective:

1. To evaluate and compare the Throughput (Mbps) performance of NB-IoT and LTE-M under varying network conditions.

2. To analyze the Latency (ms) characteristics of NB-IoT and LTE-M and study their impact on real-time communication.

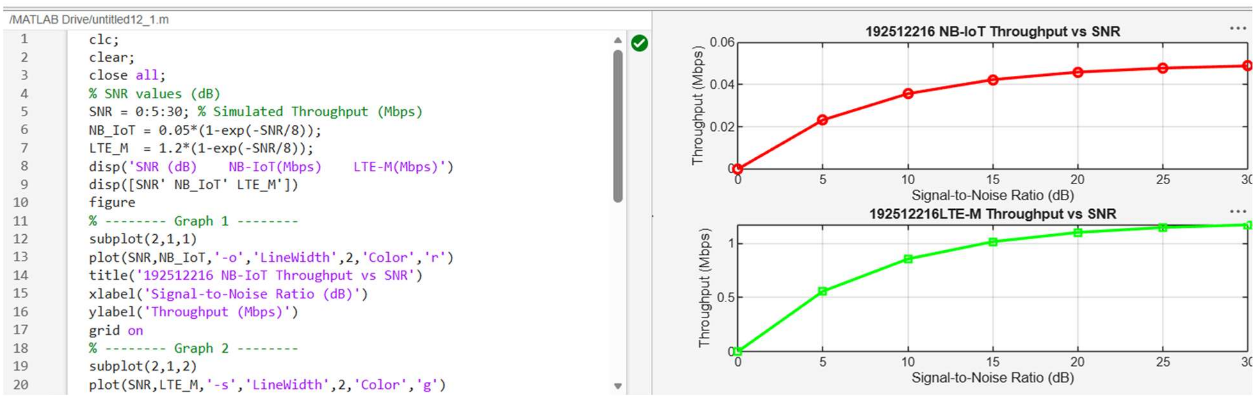
3. To measure and compare the Packet Delivery Ratio (%) of NB-IoT and LTE-M systems for reliability assessment.

Procedure:

```
clc;

clear;

close all;

% SNR values (dB)

SNR = 0:5:30; % Simulated Throughput (Mbps)

NB_IoT = 0.05*(1-exp(-SNR/8));

LTE_M  = 1.2*(1-exp(-SNR/8));

disp('SNR (dB)    NB-IoT(Mbps)    LTE-M(Mbps)')

disp([SNR' NB_IoT' LTE_M'])

figure

% -------- Graph 1 --------

subplot(2,1,1)

plot(SNR,NB_IoT,'-o','LineWidth',2)

title('NB-IoT Throughput vs SNR')

xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Throughput (Mbps)')
```

```
grid on

% -------- Graph 2 --------

subplot(2,1,2)

plot(SNR,LTE_M,'-s','LineWidth',2)

title('LTE-M Throughput vs SNR')

xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Throughput (Mbps)')

grid on
```

Output:

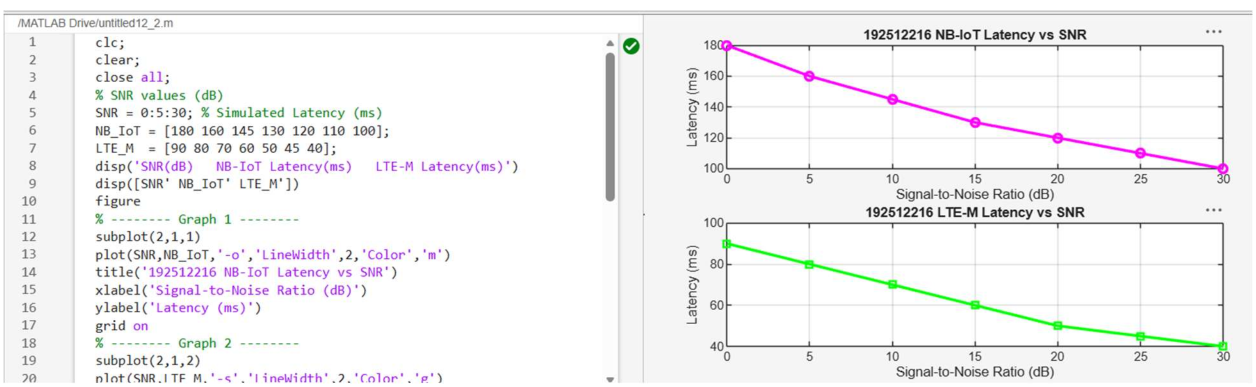


Objective 2:

```
clc;

clear;

close all;

% SNR values (dB)

SNR = 0:5:30; % Simulated Latency (ms)

NB_IoT = [180 160 145 130 120 110 100];

LTE_M  = [90 80 70 60 50 45 40];

disp('SNR(dB)   NB-IoT Latency(ms)   LTE-M Latency(ms)')

disp([SNR' NB_IoT' LTE_M'])

figure
```

```
% -------- Graph 1 --------
subplot(2,1,1)
plot(SNR,NB_IoT,'-o','LineWidth',2)
title('NB-IoT Latency vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Latency (ms)')
grid on
% -------- Graph 2 --------
subplot(2,1,2)
plot(SNR,LTE_M,'-s','LineWidth',2)
title('LTE-M Latency vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Latency (ms)')
grid on
```

Output:

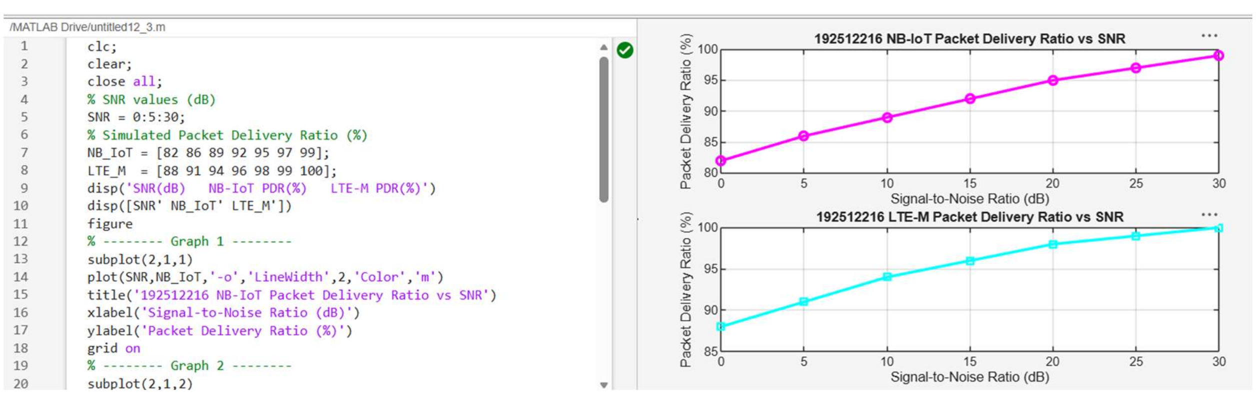


Objective 3:

```
clc;
clear;
close all;
% SNR values (dB)
SNR = 0:5:30;
```

```
% Simulated Packet Delivery Ratio (%)
NB_IoT = [82 86 89 92 95 97 99];
LTE_M  = [88 91 94 96 98 99 100];
disp('SNR(dB)   NB-IoT PDR(%)   LTE-M PDR(%)')
disp([SNR' NB_IoT' LTE_M'])
figure
% -------- Graph 1 --------
subplot(2,1,1)
plot(SNR,NB_IoT,'-o','LineWidth',2)
title('NB-IoT Packet Delivery Ratio vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Packet Delivery Ratio (%)')
grid on
% -------- Graph 2 --------
subplot(2,1,2)
plot(SNR,LTE_M,'-s','LineWidth',2)
title('LTE-M Packet Delivery Ratio vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Packet Delivery Ratio (%)')
grid on
```

Output:



Result:

1. The throughput performance of both NB-IoT and LTE-M was successfully simulated, and LTE-M achieved higher throughput than NB-IoT under the same network conditions.

2. The latency analysis showed that LTE-M provides lower communication delay compared to NB-IoT, making it more suitable for real-time IoT applications.

3. The Packet Delivery Ratio (PDR) of both technologies improved with increasing SNR, with LTE-M demonstrating slightly higher reliability than NB-IoT during data transmission.